Anirudh Kamath

Ph.D. student in Computer Science

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Education

- **University of Utah** — Salt Lake City, UT
  Doctor of Philosophy in Computer Science — *Aug. 2025 – present*
- **University of Utah** — Salt Lake City, UT
  Master of Science in Computer Science — *Aug. 2023 – Aug. 2025*
  Courses: Graduate Algorithms, Programming Languages, Advanced Computer Networks
- **Manipal Institute of Technology** — Manipal, India
  Bachelor of Technology in Computer and Communication Engineering — *Aug. 2016 – Aug. 2020*
  Courses: Machine Learning, iOS/Android Mobile App Development, Network Programming, IoT

Publications

- A. Kamath, D. Maas, J. V. d. Merwe, A. Fariha "WN-Wrangle: Wireless Network Data Wrangling Assistant," ArXiv pre-print. https://arxiv.org/abs/2603.21310.
- M. S. Hossen, C. Dickerson, O. Ozdemir, A. Gurses, M. R. Sarbudeen, T. Zajkowski, A. M. Alam, E. Tucker, W. Bjorndahl, F. Solis, S. Javed, A. Kamath, X. Tang, J. J. Sadique, M. R. Khan, K. L. Hermstein, K. A. Mahmud, J. A. S. Viloria, S. Hawkins, Y. Cui, A. Dey, Y. Liu, A. Gurbuz, J. Camp, R. Ahmad, J. V. D. Merwe, G. Zussman, M. Kurum, N. Kamesh, Z. Guan, D. Pados, G. Sklivanitis, I. Guvenc, M. Sichitiu, M. Mushi, and R. Dutta "Collection: UAV-Based Wireless Multi-Modal Measurements From AERPAW Autonomous Data Mule (AADM) Challenge in Digital Twin and Real-World Environments," in IEEE Data Descriptions, vol. 3, pp. 354-366, 2026, doi: 10.1109/IEEEDATA.2026.3685296.
- P. F. Pérez, I. Bravo, A. Kamath, C. Fiandrino and J. Widmer, "ChronoProf: Profiling Time Series Forecasters and Classifiers in Mobile Networks with Explainable AI," 2025 IEEE 26th International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM), Fort Worth, TX, USA, 2025, pp. 41-50, doi: 10.1109/WoWMoM65615.2025.00019.

Research Experience

- **University of Utah** — Salt Lake City, UT
  Wireless data discovery — *March 2026 - present*
  - Building a data extraction system that can translate a user's natural language request to a dataset containing relevant columns and experimental metadata. Research techniques used involve designing dense index structures, agentic search algorithms, and data wrangling automation.
- **University of Utah** — Salt Lake City, UT
  Wireless data wrangling — *Aug 2025 - March 2026*
  - Built a domain-specific wireless data wrangling assistant that automates wrangling multiple time-series data tables into one canonical dataset via DSL (Domain Specific Language) programmed operations to produce analysis ready wireless datasets.
- **IMDEA Networks Institute** — Madrid, Spain
  eXplainable AI systems for mobile data — *May 2024 - Aug 2024*
  - Exploring new methodologies to generate robust XAI (eXplainable AI) results for time-series forecasting models such as PatchTST, TSMixer, and DLinear. Worked with Dr. Claudio Fiandrino in the Lab for Resilient AI Networking.

Awards

- Winner of the AERPAW Autonomous Data Mule (AADM) Challenge 2025 held at North Carolina State University.
- Awarded NSF student travel grant to attend the Open6G+AI School 2026 at Northeastern University.
- Awarded NSF student travel grant to attend the AERPAW Community Workshop 2025 at North Carolina State University.

Work Experience

- **Baxter** Bengaluru, India

  Network Automation Engineer *Sep 2020 - Jul 2023*

  - Built and maintained containerized **Python** network automation applications and **REST API** services with frameworks such as **Django** hosted on **AWS Elastic Container Service** to manage network data and automate network operations for **4000+ network devices**

    Skills: ECS, EC2, CloudFormation, S3, RDS, Python, REST APIs, GraphQL, Django, PostgreSQL

  - Developed a **distributed telemetry** system that collects and stores **1M network metrics / 1.4B data points** every day from the Baxter network in a **time-series database** consumed through **Grafana** dashboards and alert notifications produced by **Alertmanager** – bringing down MTTR to a matter of minutes

    Skills: SNMP, DNS, HTTP, TLS, NGINX, Prometheus, Thanos, Telegraf, Grafana, Alertmanager

Skills

- **Wireless systems**: O-CUDU, OpenAirInterface, Open5GS, gnuradio
- **Data systems**: Apache Spark, Apache Iceberg, PostgreSQL, sqlite, InfluxDB
- **Software engineering**: Python, Ansible, Docker, Ollama
- **Software observability**: Telegraf, Prometheus, Thanos, Alertmanager, Grafana